

Research on OTFS Systems for 6G

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Abstract—The 6G communication system is expected to achieve seamless global coverage. The orthogonal time-frequency space (OTFS) is generally considered as the main candidate waveform for 6G communication. Especially, in the air-space-ground integrated communication system, OTFS is more suitable for air interface modulation waveforms for high mobility communication scenarios than OFDM system. This paper focuses on OTFS technology, which conveniently adapts to the channel constantly changing via modulating information. In the paper, comparative analysis under different rate scenarios is performed, and potential future application scenarios are proposed, such as applying artificial intelligence based on vehicle network.

Keywords—OTFS, 6G, Air-space-ground integrated, Delay-doppler domain, Artificial intelligence

I. INTRODUCTION

The 6G wireless communication network aims at providing full and multidimensional coverage for the digital and networked intelligent society. The reliability in fast-moving scenarios at high frequencies Communication is important. The OTFS system has good compatibility with the OFDM system without greatly increasing the hardware complexity, which meet the needs of the 6G waveform.

The essence of the 6G network is the extension of 5G network. The achievable conventional download speed is 1.0TB per second. The current Internet market recognizes the 6G network, and the research and development work can be carried out in 2020, and it is expected to be officially put into social application in the future.

The arrival of the 6G era has truly opened a new chapter for the construction of my country's mobile communication technology [1]. The ultimate development goal of the 6G network is to achieve the "interconnection" of "everything". 6G-oriented can not only improve the performance of mobile communication interconnection but also provide reliable technical support for Internet communication.

In the 6G era, people can not only use the Internet for online real-time online payment, image and text data transmission, and online text editing, but also some mobile game websites and live broadcast APPs have also enriched people's lives to a great extent. Compared with the traditional 5G network, the 6G network has lower latency and higher content recognition rate, which has truly realized the transition of 6G technology from satisfying personal life to satisfying social development [2].

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For 6G, the service content that can be provided to communication customers has also changed simultaneously. Facing 6G, the content of services that can be provided to communication customers has also changed simultaneously. That is to say, in the mass communication of new data in the market, we can grasp the problems encountered by different users in communication and the status of communication. And though the user's portrait, make an accurate judgment on the user's needs, to provide users with a relatively targeted service.

II. DEVELOPMENT HISTORY

Waveform Development History as shown in Figure1, Wireless communication has gone through countless ups and downs from 1G to 5G, from analog communication to digital communication, from basic voice to satisfying various business and scenario requirements [3]. The waveform design is an important factor affecting communication performance [4]. The waveform designs have also gradually changed from Frequency Division Multiple Access (FDMA) to Orthogonal Frequency Division Multiplexing (OFDM) [5].

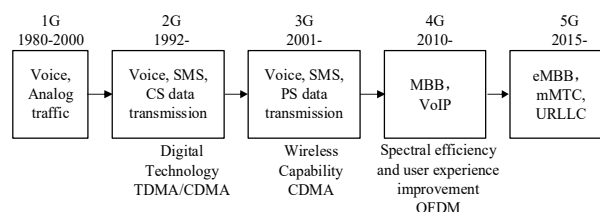


Fig. 1. Waveform development history

OFDM is the multi-carrier digital modulation such as 5G /LTE cellular and WiFi [6]. The advantages of OFDM include support for higher data rates and simpler receiver designs [7].

III. CHALLENGES FACED BY WAVEFORMS IN THE 6G ERA

LTE and 5G NR use cyclic prefix orthogonal frequency-division multiplexing (CP-OFDM)/ Discrete Fourier Transform-s-OFDM (DFT-s-OFDM) as uplink/downlink waveforms [8]. Compared with LTE and 5G NR air interface technologies, terahertz communication has abundant spectrum resources and resource advantage of ultra-large bandwidth. However, terahertz communication also faces some challenges, including high power consumption of devices, limited sampling bandwidth, high phase noise, large path loss, etc. In order to overcome these challenges, in addition to relying on the research, development, and performance improvement of high-frequency devices, efficient air interface design is also required [9]. Therefore, when designing the waveform, frame structure, modulation coding, beam management, and other technologies for the air interface technology of terahertz communication, the

irrational characteristics of the hardware link should be comprehensively considered.

6G will contain more and more complex application scenarios than 5G, and the needs of different application scenarios will be different. For some special application scenarios, in order to ensure good performance, it is very important to enhance the waveform design of the air interface [10]. At present, there is no single air interface waveform solution that can meet the needs of various 6G application scenarios. For terahertz scenarios, in order to overcome some challenges, single-carrier enhanced waveforms may be a good choice. For high Doppler shift scenarios, OTFS-based enhanced waveforms may be a good choice. For indoor hotspot coverage scenarios, in order to meet the higher rates, larger capacity needs and so on, OFDM multi-carrier-based enhanced waveforms may be a good choice [11][12]. Therefore, designing a combination scheme of multiple waveform types will meet the needs of different 6G scenarios. In the combination scheme of multiple waveform types, the flexible switching, cooperation, and compatibility between different waveforms also need to be further studied. This paper focuses on the study of OTFS technology for high Doppler frequency shift scenarios [13].

IV. OTFS DESIGN AND ANALYSIS

OTFS adds precoding unit and decoding unit to consecutive OFDM symbols, therefore, the OTFS system has good compatibility with the OFDM system without greatly increasing the hardware complexity. The principles and applications of OTFS require in-depth research and analysis.

A. Wireless channel representation

Linear time-variant (LTV) satisfies the characteristics of system superposition and uniformity. When the value of a parameter in the system changes with time, the entire characteristic also changes with time. LTV wireless channels have different representations. The transmission signal $s(t)$ is transmitted through the LTV channel, then the signal on the receiving side is $r(t)$, which is shown in (1):

$$r(t) = \int_0^{\tau_{max}} g(t, \tau) s(t - \tau) d\tau \quad (1)$$

where $g(t, \tau)$ is time-variant impulse response. In DD Channel, $r(t)$ is shown in (2):

$$r(t) = \int_{-v_{max}}^{v_{max}} \int_0^{\tau_{max}} b(\tau, v) s(t - \tau) e^{j2\pi vt} d\tau dv \quad (2)$$

where $b(\tau, v)$ is DD response (OTFS).

In time-frequency Channel, $r(t)$ as shown in (3):

$$r(t) = \int_0^{f_{max}} H(t, f) S(f) e^{j2\pi ft} df \quad (3)$$

where $H(t, f)$ is time-frequency response.

Channel in Time-Frequency $H(t, f)$ is transformed into $b(\tau, v)$ through SFFT in DD domain, otherwise, $b(\tau, v)$ through ISFFT transform to $H(t, f)$, as shown in (4) (5), This paper uses Matlab to simulate time-frequency and DD responses as shown in Figure 2-3, the main simulation parameters are presented in Table 1.

$$b(\tau, v) = \int_0^{f_{max}} \int_0^{t_{max}} H(t, f) S(f) e^{j2\pi ft} dt df \quad (4)$$

$$H(t, f) = \int_{-v_{max}}^{v_{max}} \int_0^{\tau_{max}} b(\tau, v) e^{j2\pi(vt - f\tau)} d\tau dv \quad (5)$$

TABLE I. MATLAB SIMULATION PARAMETERS

Parameter	Value
M1(resource elements in time domain)	1024
N1(resource elements in frequency domain)	128
Carrier Frequency/GHz	4
Subcarrier spacing/kHz	15
Delay spread/us	4.7
Doppler frequency/Hz	111,444,1850
Path_number	3

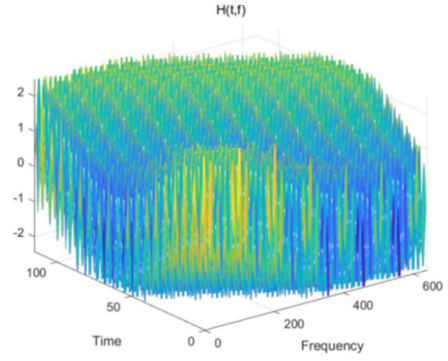


Fig. 2. Simulation of Time-Frequency

Parameters M1 and N1 are the number of resource elements in the frequency and time domain respectively.

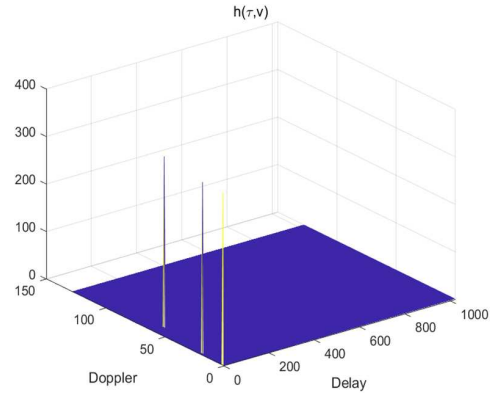


Fig. 3. Simulation of Delay-Doppler

B. OTFS modulation and de modulation

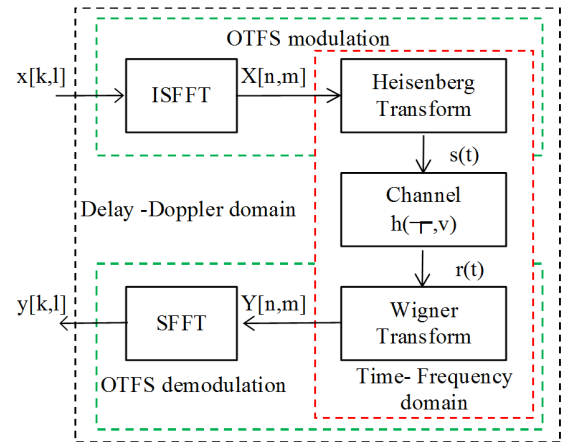


Fig. 4. OTFS modulation / demodulation

OTFS modulation /demodulation are presented in Figure 4. OTFS passes the Inverse Symplectic Finite Fourier Transform (ISFFT) maps the data symbols in the DD to the time-frequency domain and then transforms them into time-domain signals through Heisenberg Transform. For transmission through a wireless channel, the receiving end corresponds to the inverse process of the transmitting end, and the received signal in time domain is converted into the DD domain for demodulation. OTFS can be regarded as a time-frequency two-dimensional expansion technology. The ISFFT at the sending end can be implemented respectively by performing M-point discrete Fourier transform (DFT) and N-point inverse discrete Fourier transform (IDFT) on the columns and rows of the DD domain signal matrix. The receiving section converts the received signal from the time domain to the time-frequency domain through the Wigner transform and then transforms from the time-frequency to the DD domain through the Symplectic Finite Fourier Transform(SFFT)[14][15].

In the Time-Frequency domain, The modulator as Heisenberg transform as shown in (6), from $r(t)$ simplifies to the discrete Finite Fourier Transform (IFFT) in the case of $N=1$ and rectangular, we can see that matched filter as Wigner transform as shown in (7)-(8).

$$s(t) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X[n, m] b_{tx}(t - nT) e^{j2\pi m \Delta f (t - nT)} \quad (6)$$

$$Y(t, f) = \int_0^{t'} \max b_{rx}^*(t' - t) r(t') e^{-j2\pi f(t' - t)} dt' \quad (7)$$

$$Y[n, m] = Y(t, f)|_{t=nT, f=m\Delta f} \quad (8)$$

where parameters n and m are represent the index of the time-frequency domain resource grid point on the time and frequency axis, nT is the duration time, $m\Delta f$ is the occupied bandwidth, b_{tx} is send pulse, $b_{tx}(t)$ and $y(t)$ are fuzzy functions of each other, $Y(t, f)$ is cross blur function.

C. Design and Analysis of Channel Estimation in OTFS System

When the signal is transmitted through the wireless channel, the amplitude, frequency, and phase will change due to the influence of obstacles. The received signal must be channel estimated to obtain the correct received signal information.

(1) Ideal pulse signal

In the Time-Frequency domain ideal pulses, if $b_{rx}(t)$ and $b_{tx}(t)$ perfectly localize in frequency and time, and $H[n, m]$ and $Y[n, m]$ relationship satisfaction in (9)-(10).

$$Y[n, m] = H[n, m] X[n, m] \quad (9)$$

$$H[n, m] = \int_{-v_{max}}^{v_{max}} \int_0^{\tau_{max}} h(\tau, v) e^{j2\pi v n T} e^{-j2\pi m \Delta f \tau} d\tau dv \quad (10)$$

Simulation parameters as show in Table2, $X[n, m]$ and $Y[n, m]$ stands for transmitting and receiving side of signal, $H[n, m]$ is window function, $\text{abs}(w)$ stands for w 's absolute value or complex modulus. By default, ymbolic variables are assumed to be complex, 'abs' return the complex modulus (magnitude). If z is an array, abs acts element-wise on each element of w .

Simulation of transmitting and receiving side of signal 'abs' is shown in Figure5 and Figure6, the horizontal axis is the Doppler unit is Hz, the vertical axis is the Delay unit is 10us. Specifically, data symbols are placed in purple at the transmitting side, pilot signals are placed in other areas, data

symbols are placed in purple at the receiving side for detection, and channel estimation is placed in the remaining areas.

TABLE II. SIMULATION PROGRAM

Main Parameter Program
dd/xmit = zeros(N,M)
X = fftshift(fft(fftshift(ifft(dd_xmit),1),2))
r = zeros(size(s))
r_mat = reshape(r,N,M)
Y = fft(r_mat,M,1)/sqrt(M)
dd/rx = fftshift(ifft(fftshift(fft(Y),1)))

TABLE III. SIMULATION PARAMETERS FOR IDEAL PULSE SIGNAL

Parameter	Value
span of delay axis/us	50
span of Doppler axis /kHz	20
number of cells along Doppler	20
Number of cells across Delay	20
SNR /dB	30

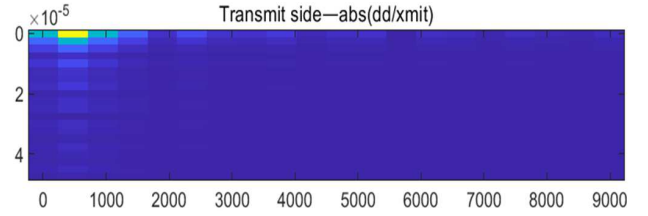


Fig. 5. Simulation of transmitting side of signal 'abs'

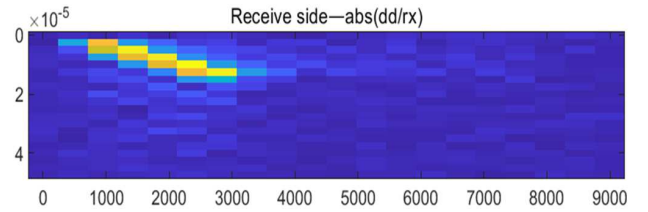


Fig. 6. Simulation of receiving side of signal 'abs'

(2) Doppler domain channel estimation scheme

The input of the OTFS system is the data symbol $x[k, l]$ after digital modulation (such as 4QAM modulation), $x[k, l] \in A$. The system output is $y[l_1, l_2]$, $l_1 = \{0, \dots, N-1\}$, $l_2 = \{0, \dots, M-1\}$, the OTFS system output $Y[n, m]$ in the DD domain as shown in (11).

$$H[n, m] = \sum_{l_1=0}^{N-1} \sum_{l_2=0}^{M-1} h[l_1, l_2] e^{j2\pi(\frac{nl_1}{N} - \frac{ml_2}{M})} \quad (11)$$

where parameters l_2 and l_1 represent the index of DD resource grid point on the delay and Doppler axis, $X[n, m]$, $y[l_1, l_2]$ are ISFFT and SFFT operations for OTFS modulation, as shown in (12) (13). After ISFFT as shown in (12), the corresponding SFFT as shown in (13).

$$X[n, m] = \frac{1}{\sqrt{NM}} \sum_{l_1=0}^{N-1} \sum_{l_2=0}^{M-1} x[l_1, l_2] e^{j2\pi(\frac{nl_1}{N} - \frac{ml_2}{M})} \quad (12)$$

$$y[l_1, l_2] = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y[n, m] e^{-j2\pi(\frac{nl_1}{N} - \frac{ml_2}{M})} \quad (13)$$

DD domain input and output relation as shown in (14).

$$y[l_1, l_2] = \sum_{i=1}^P h_i e^{-j2\pi\tau_i v_i} x[[l_1 - l_{1i}]_N, [l_2 - l_{2i}]_M] \quad (14)$$

The input and output relationship can also be represented by a two-dimensional matrix $Y = H_{dd}X$, as shown in (15).

$$y[l_1, l_2] = h[l_1, l_2] \odot x[l_1, l_2] \quad (15)$$

Doppler domain channel estimation scheme, simulation of transmitting and receiving side of signal 'abs as shown in Figure 7-8, in the OTFS system, if the delay and Doppler resolution are strong enough, the channel presents several sparse impulses in the DD domain, such as impulses with multiple guard interval symbols around the pilot in the common DD domain are presented in Figure 7. The ordinate is the delay (M), and the abscissa is the doppler (N). In Figure 7, the purple display is the guard interval, the purple middle is the pilot, and the yellow is the data. In Figure 8, the blue continuous part of the receiving side is the channel estimation, and the rest is the detection of data.

TABLE IV. SIMULATION PARAMETERS FOR DOPPLER DOMAIN

Parameter	Value
span of delay axis /us	50
span of Doppler axis /kHz	20
number of cells along Doppler	20
Number of cells across Delay	20
SNR/dB	30
Modulation	4QAM

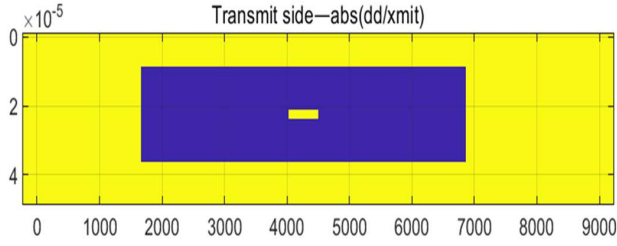


Fig. 7. Simulation of transmitting side of signal 'abs

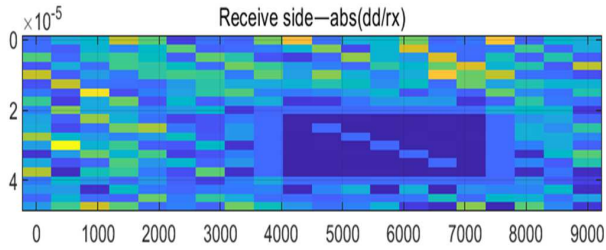


Fig. 8. Simulation of receiving side of signal 'abs

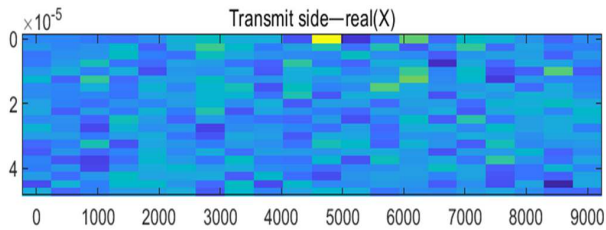


Fig. 9. Simulation of transmitting side of signal 'abs

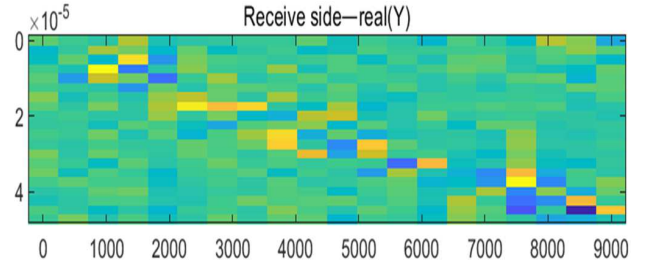


Fig. 10. Simulation of receiving side of signal 'abs

Data symbols are placed in blue at the transmitting side pilot signals are placed in the remaining areas, data symbol detection is placed in green at the receiving side, and channel estimation is placed in the remaining areas as shown in Figure 9-10. Figure 7-10 The horizontal axis is the Doppler unit is Hz, the vertical axis is the Delay unit is 10us.

In the DD domain data frame, the pilot frequency indicated by $x[k, l]$ is inserted to estimate the wireless channel. On the receiving side, the received OTFS symbol is the DD domain channel, and the two-dimensional convolution of the DD domain is shown in $y[k, l]$. Use bar3 (Z) in Matlab to draw a 3D bar graph, each element in Z corresponds to a bar graph. If Z is a vector, the y-axis scales from 1 to length (Z). If Z is a matrix, the scale of the y-axis is from 1 to the number of rows in Z. Input $x[k, l]$ as shown in Figure 11, output $y[k, l]$ as shown in Figure 12. The X and Y axis coordinates of the Figure 11-12 is number of cells along Doppler, Number of cells across Delay, unit is number of cells.

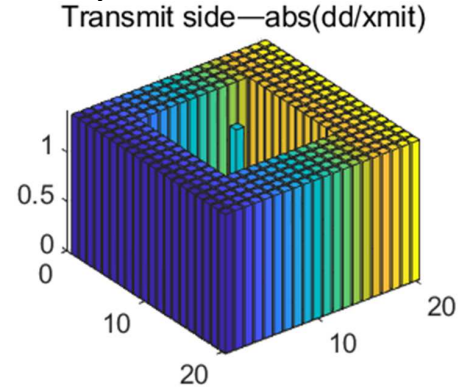


Fig. 11. Simulation of transmitting side of signal 'abs

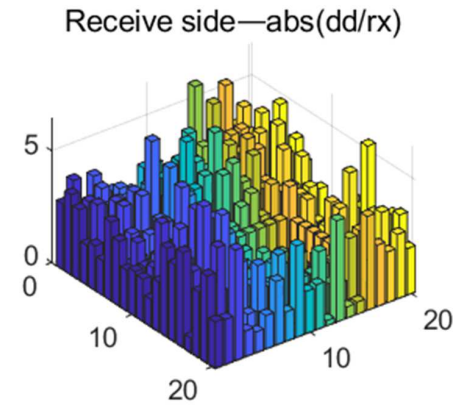


Fig. 12. Simulation of receiving side of signal 'abs

D. Comparison of OFDM and OTFS

It is generally known that the control channel's coding scheme is the Polar code. The data channel's coding scheme is the LDPC code [16] [17]. Channel encoding and decoding is the improvement of its performance will directly improve network coverage and user transmission rate [18]. As a fundamental innovation in the field of channel encoding and decoding, the introduction of Polar code will significantly improve the user experience of 5G networks and further enhance the competitiveness of 5G standards. BER for 4-QAM in Wideband Rayleigh Fading as shown in Figure 13, when the SNR is the same, the OTFS' BER is much lower than OFDM's.

TABLE V. SIMULATION PARAMETERS

Parameter	Value
UE speed /km/h	120
code rate	2/4
Set maximum no. of iterations for LDPC decoder	25
Number of simulation repetitions	1

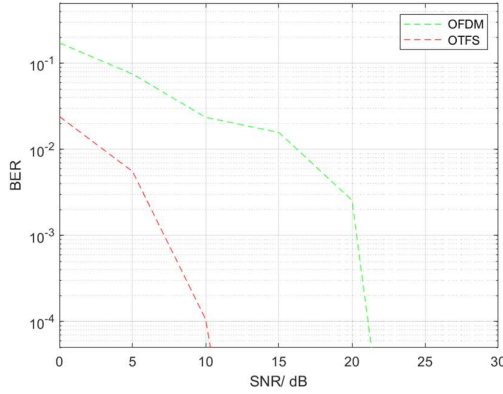


Fig. 13. BER simulation curve of OFDM and OTFS

E. Comparison of OTFS differences at different rates

To evaluate OTFS system performance, this part obtains the bit error rate (BER) of channel estimation through Matlab simulation. The simulation of OTFS performance at different rates is shown in Figure 14. In the mobility scenario, when the SNR is the same, the BER of OTFS modulation is hardly affected by the Doppler frequency offset.

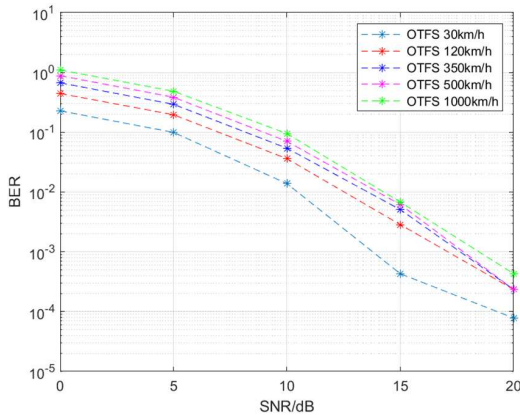


Fig. 14. BER simulation curve of OTFS (different UE speed)

V. POTENTIAL APPLICATION SCENARIOS

In the 6G network, OTFS will be related to various services and applications. OTFS will exert its value as described below.

A. Millimeter-wave Communications

As we all know, in order to reduce ICI the cyclic prefix (CP) can be inserted in the guard interval. The sample points after each OFDM symbol are copied to the front of the OFDM symbol, which ensures that the number of waveform periods contained in the delayed duplicate of the OFDM symbol is an integer number in the FFT period, and the orthogonality of the subcarriers is guaranteed [19].

Inter-Symbol Interference (ISI) seriously affects digital signals' transmission quality. Inter-Channel Interference (ICI) is when the OFDM sub-carriers' orthogonality is destroyed, thereby affecting the receiving end's demodulation [20].

The mmWave spectrum holds a lot of underutilized spectrum and has the potential ability to transfer quickly in 6G networks. OTFS is highly immune to oscillator phase noise, which is critical for mmWave communications [21].

B. Satellite channel

Satellite channels are similar to terrestrial channels and belong to wireless channels, but due to their special environment, satellite channels have unique channel characteristics. Satellite channels have the following characteristics [22][23]:

(1) Signals will produce multipath effects due to scattering, reflection, and diffraction during transmission;

(2) Due to multipath effects and the movement of satellites and ground terminals, satellite-ground channels belong to frequency selective fading and time selective fading channels (dual selective channel);

(3) The signal is mostly the line of sight (LOS) component, which follows Rician distribution, which has less reflection component. The shadow effect is caused by the occlusion of trees or buildings. The rain attenuation effect is caused by atmospheric conditions such as rain, snow, clouds, and fog [24].

The application of OTFS in satellite communication has the advantages of solving the signal transmission problem of broadband satellite-ground link in high moving speed scenarios, which is beneficial to the communication between satellite and ground [25].

C. Radar Communications

The radar estimates the distance and speed of the target according to the time delay and Doppler frequency in the echo to achieve target detection, which is similar to the equivalent time delay Doppler channel constructed by OTFS [26].

D. Artificial Intelligence based on vehicle network

After using the OTFS, the transmission performance is significantly improved in high-speed mobile scenarios, and it has higher spectrum utilization and lower peak to average power ratio (PAPR), OTFS is more suitable for high mobility communication. OTFS can play a vital role in 6G network as it is superior to high mobility channels for communication. The combination of AI-based in-vehicle and OTFS helps to provide better network coverage and intelligent control management for in-vehicle networks.

VI. CONCLUSION

OFDM is very sensitive to frequency offset, and the Doppler frequency shift caused by high-speed movement between satellite and ground will degrade its performance [27]-[29]. As a new carrier modulation technology, OTFS can combat the non-ideal characteristics of satellite-to-ground link transmission to improve system transmission. This paper firstly replies to the history of communication development, and the challenges faced by waveforms in the 6G era analyzed. Then the OTFS concepts and features are elaborated, including the main features in DD domain, etc. It then highlights the key challenges of OTFS. Finally, the potential applications of OTFS and several promising research directions are introduced [30]-[32]. Meanwhile, we also see that OTFS still needs to further development, such as design complex beamforming and precoding to take full advantage of all available spatial degrees of freedom, and perform complexity detection for MIMO-OTFS, etc.

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